



Q.1 (D)

Roots are $\alpha, 6\alpha$

$$\alpha + 6\alpha = 7 \sin^{-1} x \Rightarrow \alpha = \sin^{-1} x$$

$$(6\alpha)\alpha = 12\pi \sin^{-1} x + f(x)$$

$$6(\sin^{-1} x)^2 = 12\pi \sin^{-1} x + f(x)$$

$$f(x) = 6[(\sin^{-1} x)^2 - 2\pi \sin^{-1} x]$$

$$f(x) = 6[(\sin^{-1} x - \pi)^2 - \pi^2]$$

$$\text{minimum value of } f(x) \text{ is } \frac{-9\pi^2}{2} \text{ when } \sin^{-1} x = \frac{\pi}{2}$$

$$\text{Maximum value of } f(x) \text{ is } \frac{15\pi^2}{2} \text{ when } \sin^{-1} x = \frac{-\pi}{2}$$

$$\Rightarrow \frac{2(M+m)}{\pi^2} = 6. \text{ Ans.}$$

Q.2 (A)

$$f^2(-1) + f^2(0) + f^2(1) + 30 = 4f(0) + 10f(1) + 2f(-1) \text{ can be written as}$$

$$(f^2(-1) - 2f(-1) + 1) + (f^2(0) - 4f(0) + 4) + (f^2(1) - 10f(1) + 25) = 0$$

$$(f(-1) - 1)^2 + (f(0) - 2)^2 + (f(1) - 5)^2 = 0$$

$$\Rightarrow f(-1) = 1, f(0) = 2, f(1) = 5 \Rightarrow f(x) = x^2 + 2x + 2 = (x+1)^2 + 1$$

$$\Rightarrow 1 \leq f(x) < \infty. \text{ Ans.}$$

Q.3 (D)

$$\cos(\cos^{-1} \pi x) \text{ is periodic with period : } \frac{2\pi}{\pi} = 2$$

$$|\sin \pi x| \text{ is periodic with period : } \frac{\pi}{\pi} = 1$$

$$x^2 - 10x + 30 > 0 \quad (D < 0, a > 0)$$

$$\Rightarrow \text{sgn}(x^2 - 10x + 30) = 1$$

$$\sin n\pi = 0 \Rightarrow \sin \pi [x] = 0$$

$$\text{Period is LCM of } \{2, 1\} = 2. \text{ Ans.}$$

Q.4 (C)

Replace x by $(1-x)$, we get

$$\frac{f^2(1-x)}{f(x)} = (1-x)^3$$

$$\therefore f^3(x) = x^6 (1-x)^3 \Rightarrow f(x) = x^2 (1-x) \Rightarrow f\left(\frac{1}{2}\right) = \frac{1}{2} \text{ Ans.}$$



Q.5 (C)

For $f(x)$ to be even

We must have

$$[\lambda]^3 - 9[\lambda]^2 + 26[\lambda] - 24 = 0 \Rightarrow [\lambda] = 2, 3, 4$$

$$\text{and } 24\{\lambda\}^3 - 26\{\lambda\}^2 + 9\{\lambda\} - 1 \Rightarrow \{\lambda\} = \frac{1}{2}, \frac{1}{3}, \frac{1}{4}$$

$$\Rightarrow \lambda \text{ can be } 2 + \frac{1}{2}, 2 + \frac{1}{3}, 2 + \frac{1}{4}, 3 + \frac{1}{2}, 3 + \frac{1}{3}, 3 + \frac{1}{4}, 4 + \frac{1}{2}, 4 + \frac{1}{3}, 4 + \frac{1}{4}$$

$$\begin{aligned} \text{Sum} &= \left(\frac{5}{2} + \frac{7}{3} + \frac{9}{4}\right) + \left(\frac{7}{2} + \frac{10}{3} + \frac{13}{4}\right) + \left(\frac{9}{2} + \frac{13}{3} + \frac{17}{4}\right) = \frac{21}{2} + 10 + \frac{39}{4} \\ &= \frac{42 + 40 + 39}{4} = \frac{121}{4} \text{ Ans.} \end{aligned}$$

Q.6 (C)

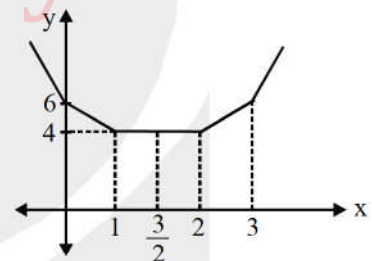
$$f(x) = |x| |x-1| |x-2| |x-3| \quad \forall x \in [1, 2]$$

$$f\left(\frac{3}{2}\right) = \frac{9}{16}, \quad f\left(\frac{4}{3}\right) = \frac{20}{81}, \quad f\left(\frac{5}{3}\right) = \frac{20}{81}$$

$$f(x) \text{ is symmetric about } x = \frac{3}{2}$$

$$\text{and } f(x)_{\max} = \frac{9}{16} \text{ at } x_0 = \frac{3}{2}$$

$$g(x) = |x| + |x-1| + |x-2| + |x-3| = \begin{cases} -4x+6; & x \leq 0 \\ -2x+6; & x \in (0, 1) \\ 4; & x \in [1, 2] \\ 2x; & x \in (2, 3) \\ 4x-6; & x \geq 3 \end{cases}$$



$$g(x) \text{ is minimum at } x_0 = \frac{3}{2} \text{ and its value is 4.}$$

$$\text{Given } f(x)_{\max} = \lambda g(x)_{\min} \text{ at } x = x_0 \text{ where } x_0 = \frac{3}{2}$$

$$\Rightarrow \frac{9}{16} = \lambda (4) \Rightarrow \lambda = \frac{9}{64}$$

$$\therefore \lambda + x_0 = \frac{9}{64} + \frac{3}{2} = \frac{105}{64} \text{ Ans.}$$

Q.7 (B)

$$\alpha = \frac{2}{3}; \quad \beta = \frac{1}{3}.$$

Hence the quadratic equation whose roots are α and β .

$$x^2 - x + \frac{2}{9} = 0 \Rightarrow 9x^2 - 9x + 2 = 0$$

$$a + b + c = 2 \text{ Ans.}$$

Q.8 (A)

$$\therefore |f(1) + 3| < |f(1)| + 3 \Rightarrow f(1) < 0$$

$$\text{and } |f(2) + 10| = |f(2)| + 10 \Rightarrow f(2) > 0$$

$\therefore f(x)$ will have atleast one real root in $(0, 1)$.

Q.9 (A) -2 (B) -5 (C) 2 (D) 5

Q.10 (C)

Q.11 (C)

$$\therefore f(2) - f(1) = -2$$

$$\text{and } f(3) - f(2) = -3$$

$$\therefore f(x+1) - f(x) = -(x+1) \text{ for } x = 1 \text{ and } 2 \text{ only}$$

$$\Rightarrow f(x+1) - f(x) = (x+1)^3 + a(x+1)^2 + b(x+1) + c - (x^3 + ax^2 + bx + c)$$

$$= x^3 + 3x^2 + 3x + 1 + a(x^2 + 2x + 1) + bx + b + c - x^3 - ax^2 - bx - c$$

$$= 3x^2 + (3 + 2a)x + a + b + 1$$

$$\therefore 3x^2 + (3 + 2a)x + (a + b + 1) + (x + 1) = 0 \text{ will have roots } x = 1 \text{ and } 2$$

$$= 3(x-1)(x-2)$$

$$= 3(x^2 - 3x + 2)$$

$$\therefore 4 + 2a = -9 \Rightarrow a = \frac{-13}{2}$$

$$\text{and } a + b + 2 = 6 \Rightarrow a + b = 4 \Rightarrow b = \frac{21}{2}$$

$$\therefore f(x+1) - f(x) = 3(x-1)(x-2) - (x+1)$$

Putting $x = 3$

$$f(4) - f(3) = 3 \times 2 \times 1 - 4 = 2 \Rightarrow f(3) = f(4) - 2$$

$$\therefore f(x) = x^3 - \frac{13}{2}x^2 + \frac{21}{2}x + C$$

$\therefore f(x)$ is cubic, $\therefore$ onto function

$$f'(x) = 3x^2 - 13x + \frac{21}{2}$$

$$\therefore D = (13)^2 - 4 \cdot 3 \cdot \frac{21}{2} = 169 - 6 \cdot 21 > 0$$

$\therefore f(x)$ is many one

$\therefore f(x)$ is many one and onto function.

Q.12 (D)

Q.13 (A)

Q.14 (A)

$$\text{Let } f(x) = 2x^2 - 2(\tan^{-1}\alpha - \cot^{-1}\alpha)x - (\tan^{-1}\alpha \cot^{-1}\alpha) = Ax^2 + Bx + C$$

$$\Rightarrow A = 2, B = -2(\tan^{-1}\alpha - \cot^{-1}\alpha), C = -(\tan^{-1}\alpha)(\cot^{-1}\alpha)$$



$$D = B^2 - 4AC = 4[\tan^{-1}\alpha - \cot^{-1}\alpha]^2 - 4(2)(-1)(\tan^{-1}\alpha)(\cot^{-1}\alpha)$$

$$D = 4[(\tan^{-1}\alpha)^2 + (\cot^{-1}\alpha)^2] \Rightarrow D > 0$$

$$D = 4[(\tan^{-1}\alpha + \cot^{-1}\alpha)^2 - 2\tan^{-1}\alpha \cot^{-1}\alpha]$$

$$D = 4\left[\frac{\pi^2}{4} - 2(\tan^{-1}\alpha)\left(\frac{\pi}{2} - \tan^{-1}\alpha\right)\right]$$

$$D = 4\left[2\left(\tan^{-1}\alpha - \frac{\pi}{4}\right)^2 + \frac{\pi^2}{8}\right]$$

$$\frac{\pi^2}{2} \leq D \leq \frac{5\pi^2}{2}$$

$$\text{We know vertex is } \left(\frac{-B}{2A}, \frac{-D}{4A}\right) = (h, k)$$

$$\Rightarrow h = \frac{-B}{2A}$$

$$h = \frac{\tan^{-1}\alpha - \cot^{-1}\alpha}{2}$$

$$h \leq 0 \text{ because } \alpha \in [-1, 1]$$

$$[\text{Note : } h = 0 \Rightarrow \tan^{-1}\alpha = \cot^{-1}\alpha \Rightarrow \alpha = 1]$$

$$k = \frac{-1}{4 \times 2}(D)$$

$$k = \frac{-D}{8} \Rightarrow \frac{-5\pi^2}{16} \leq k \leq \frac{-\pi^2}{16}$$

$$-5 \leq \frac{16k}{\pi^2} \leq -1$$

$$\text{i.e. } \frac{16k}{\pi^2} \text{ can take 5 integral values.}$$

Q.15 (AC)

$$\therefore f^2(x) \cdot f\left(\frac{1-x}{1+x}\right) = x^3 \quad \dots\dots(1)$$

Replace x by $\left(\frac{1-x}{1+x}\right)$, we get

$$f^2\left(\frac{1-x}{1+x}\right) f(x) = \left(\frac{1-x}{1+x}\right)^3 \quad \dots\dots(2)$$

From $\frac{(1)^2}{(2)}$, we get

$$f^3(x) = x^6 \left(\frac{1+x}{1-x}\right)^3$$

$$\therefore f(x) = x^2 \left(\frac{1+x}{1-x} \right)$$

$$\therefore f(2) = 4 \left(\frac{3}{-1} \right) = -12$$

$$\text{and } f(-2) = 4 \left(\frac{-1}{3} \right) = -\frac{4}{3}$$

$$f(2)f(-2) = 16$$

$$\text{and } f(3) = \frac{9 \times 4}{-2} = -18 \text{ Ans.}$$

Q.16 (BC)

Domain of $y = f(f(x))$

$$\text{Let } f(x) = t \Rightarrow t \in [0, 5]$$

$$y = f(t)$$

But domain of f is $[-4, 3]$

$$\therefore t \in [-4, 3] \Rightarrow f(x) \in [-4, 3] \Rightarrow x \in \left[-3, \frac{13}{5} \right]$$

Number of solution(s) of the equation $f(f(x)) = 1$

$$f(t) = 1, f(x) = t$$

↓

$$t = -1, 0, \frac{3}{2}, \frac{11}{5}$$

Now, $f(x) = -1 \Rightarrow$ No solution

$f(x) = 0 \Rightarrow$ 2 solutions

$f(x) = \frac{3}{2} \Rightarrow$ 3 solutions

$f(x) = \frac{11}{5} \Rightarrow$ 2 solutions

Hence, $f(f(x)) = 1$ has exactly 7 solutions.

Q.17 (ABCD)

8, 18, 32, 50, are the member of a series whose n^{th} term is $2n^2 + 4n + 2$ (using method of difference)

$$T_n = \tan^{-1} \frac{1}{2n^2 + 4n + 2} = \tan^{-1} \frac{2}{4n^2 + 8n + 4} = \tan^{-1} \frac{2}{1 + 4n^2 + 8n + 3}$$

$$= \tan^{-1} \frac{2}{1 + (2n+1)(2n+3)} = \tan^{-1} \frac{(2n+3) - (2n+1)}{1 + (2n+1)(2n+3)}$$

$$T_n = \tan^{-1} (2n+3) - \tan^{-1} (2n+1)$$

$$\text{Hence the summation is } \frac{\pi}{2} - \tan^{-1} 3 = \cot^{-1} 3 = \tan^{-1} \frac{1}{3} = \theta$$

$$\sin 2\theta = \frac{2 \tan \theta}{1 + \tan^2 \theta} = \frac{3}{5}; \quad \cos 2\theta = \frac{1 - \tan^2 \theta}{1 + \cos^2 \theta} = \frac{4}{5}.$$

Q.18 (A) P, T, R ; (B) P, R, T ; (C) P ; (D) P, Q

$$25 \left((x-2)^2 + (y-3)^2 \right) = C^2 (3x - 4y + 7)^2$$

$$\Rightarrow \sqrt{(x-2)^2 + (y-3)^2} = |C| \frac{|3x - 4y + 7|}{\sqrt{3^2 + (-4)^2}}$$

PF = |C| PN where F is (2, 3) and line is $3x - 4y + 7 = 0$.

We know

$|\sin^{-1} \lambda| = 0, 1, 2 \Rightarrow P$ can be a on parabola, hyperbola or point F.

$|\cot^{-1} \lambda| = 0, 1, 2, 3 \Rightarrow P$ can be on parabola, point, hyperbola.

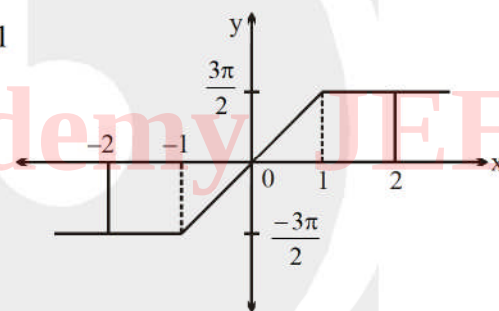
$|\cos \lambda| = 0, 1 \Rightarrow P$ can be on parabola.

$|\lambda^2| = 0, 1, 2, 3, \dots \Rightarrow P$ can be ellipse, parabola.]

Q.19 7

$$f(x) = \begin{cases} \frac{3}{2}(\pi - 2 \tan^{-1} x) + 3 \tan^{-1} x = \frac{3\pi}{2}; & x > 1 \\ \frac{3\pi}{2} x; & x \in [-1, 1] \\ -2 \tan^{-1} x + 2 \tan^{-1} x - \frac{3\pi}{2} = \frac{-3\pi}{2}; & x < -1 \end{cases}$$

$$\begin{aligned} \text{Area} &= \left(1 \times \frac{3\pi}{2}\right) 2 + \left(\frac{1}{2} \times 1 \cdot \frac{3\pi}{2}\right) 2 \\ &= \frac{6\pi}{2} + \frac{3\pi}{2} = \frac{9\pi}{2} = \left(\frac{9}{2}\right) \pi \end{aligned}$$

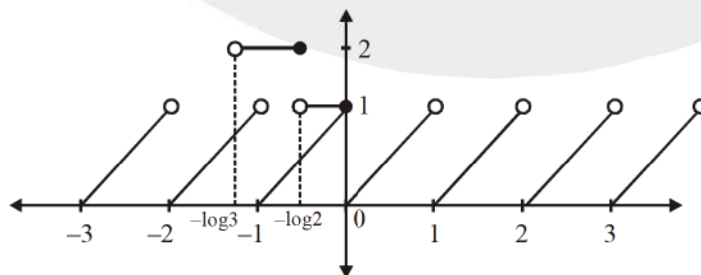


$$p = 9; q = 2$$

$$\therefore p - q = 7 \text{ Ans.}$$

Q.20 5050

$$[e^{-x}] = \begin{cases} 0, & \text{when } x > 0 \\ \text{greater than or equal to 1, when } x \leq 0 \end{cases}$$



Hence, solution are 1, 2, 3,, 100.

$\therefore$ Sum is 5050. Ans.



Q.21 4

Let $\sqrt{x-1} + \sqrt{17-x} = t$

$$\Rightarrow x-1+17-x+2\sqrt{64-(x-9)^2} = t^2$$

$$\Rightarrow 16+2\sqrt{64-(x-9)^2} = t^2$$

$$\Rightarrow f(t) = \sqrt{4+t^2}, \quad t \in [4, 4\sqrt{2}]$$

$$f(t) = 5 \Rightarrow 4+t^2 = 25 \Rightarrow t = \sqrt{21}$$

$$\Rightarrow f^{-1}(5) = \sqrt{21} \Rightarrow [f^{-1}(5)] = 4$$

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